

Balakrishna Nair Nalivalapil

College Park, MD (Open to Relocation) | (240) 476-6045 | bkrishna@terpmail.umd.edu
linkedin.com/in/bkrishnanair | github.com/bkrishnanair | bkrishnanair.github.io

Professional Summary

Software Engineer with 2 years of enterprise back-end experience and a Master's in Software Engineering from the University of Maryland (May 2026). Engineered production RESTful services and integration pipelines for MUFG Bank's \$100M+/day transaction platform at Capgemini, and independently built and launched Huddle, a production geospatial platform serving live users. Core strengths: Java, Python, TypeScript, microservices, LLM API integration, automated testing, and CI/CD on AWS and GCP.

Technical Skills

- Languages:** Java, Python, TypeScript, JavaScript, SQL, C++
Back-End & Systems: REST APIs, Microservices, Event-Driven Architecture (WebSphere MQ, IBM Integration Bus), Spring Boot, Node.js, Django, Distributed Systems, Data Structures & Algorithms
Front-End: React, Next.js, Tailwind CSS, HTML5/CSS3, Responsive Design
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), GCP, Firebase, Docker, Jenkins, JFrog Artifactory, CI/CD, Git, GitHub Actions, Linux/Unix
Databases: PostgreSQL, MongoDB, Firestore, MySQL, Redis
AI & LLM Integration: LLM APIs (Gemini, Llama3), Prompt Engineering, Agent Workflows, AI Coding Assistants (GitHub Copilot, Cursor), Pandas
Testing & Practices: Unit Testing (JUnit, PyTest, Jest), Integration Testing, TDD, Code Review, Agile/Scrum, Technical Documentation

Professional Experience

- Capgemini Technology Services** Oct 2022 – Sep 2024
Enterprise Integration: Financial Services (MUFG Bank) Chennai, India
- Senior Software Engineer** Apr 2024 – Sep 2024
- Reduced post-deployment incidents 25% by building event-driven middleware with structured logging and monitoring across distributed services using WebSphere MQ.
 - Reduced onboarding ramp-up 15% by authoring architectural decision records and mentoring 3 junior engineers on team coding standards and review practices.
- Software Engineer** Oct 2022 – Mar 2024
- Engineered 20+ production RESTful services and integration pipelines for MUFG Bank's transaction platform processing \$100M+ daily, designing Java-based middleware on IBM Integration Bus within a 100+ member global engineering organization.
 - Cut feature integration time 30% across 5 application teams by architecting automated build-test-deploy CI/CD pipelines with Jenkins and JFrog Artifactory.
 - Delivered features in two-week Agile sprints, authoring and reviewing pull requests against established coding standards and linting rules.
 - Guaranteed data integrity across heterogeneous banking systems by designing validation and transformation layers handling JSON/XML payloads at high message volume.

Projects

- Huddle | Founder & Lead Engineer | Next.js 15, TypeScript, React, Firebase, Gemini API, Vercel** Aug 2025 – Present
- Architected and shipped a production geospatial platform end-to-end with 48 API routes, 68 pages, real-time Firestore listeners, and geohash-indexed map discovery serving 565+ campus events.
 - Eliminated RSVP race conditions by consolidating capacity checks, waitlist promotions, and badge awards into a single atomic Firestore transaction under concurrent load.
 - Cut public API payload size by 87% (11 KB to 1.5 KB per event) via field allowlists, and migrated 566 live attendee records to access-controlled subcollections with a zero-data-loss verification script.
 - Consolidated 5 background cron jobs into a fault-isolated dispatcher, eliminated full collection scans, and hardened Firestore security rules via emulator-based tests and CI gating.
- ClassNest: AI-Powered LMS | Python, Django, React, PostgreSQL, Llama3, Docker** Sep 2024 – Nov 2024
- Automated PDF-to-quiz conversion by engineering an LLM pipeline with Llama3 via the Groq API; containerized with Docker and shipped through CI/CD.
 - Enforced 90% test coverage through Test-Driven Development, writing unit and integration tests across Django services and React components.
- IoT Smart Health Monitoring System | Python, Arduino, Raspberry Pi, Sensors** 2021 – 2022
- Enabled remote patient vitals tracking with threshold-based alerts by integrating 5 hardware sensors into a real-time monitoring dashboard.

Education

- University of Maryland, College Park** May 2026
Master of Engineering in Software Engineering (GPA: 3.7/4.0) College Park, MD
- Coursework:** Cloud Computing, Cloud Security, Software Design & Architecture, Virtualization & Containers, Software Project Management
- Indian Institute of Information Technology, Sri City** May 2022
Bachelor of Technology in Electronics & Communications Engineering Sri City, India

Certifications

- AWS Certified Solutions Architect: Associate (SAA-C03)** (in progress)